

	$t - 20$	$t - 15$	$t - 10$	$t - 5$	t	causal	$\hat{\sigma}$
Senseiver-A		yes	current second only				
Senseiver-G (ours)		yes	last 16 s				
DINCAE		no	✓	one second either side			
4DVarNet		no		fixed 200-s tile containing t			
4DVarNet (aug. head)		no	✓	fixed 200-s tile containing t			
EnKF		yes	✓	all past, carried by the ensemble			
observations used reconstructed frame t							